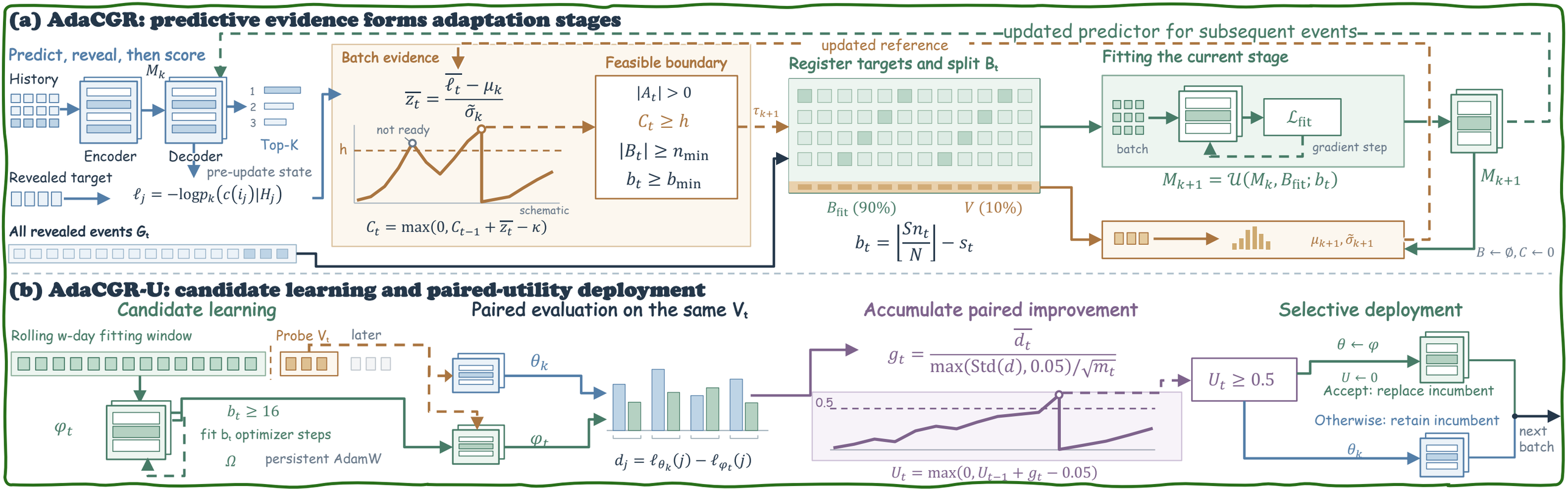

(a) AdaCGR: predictive evidence forms adaptation stages
updated predictor for subsequent events
Predict, reveal, then score
History
M_k
Encoder
Decoder
1
2
3
Top-K
pre-update state
Revealed target
$\ell_j = -\log p_k(c(i_j)|H_j)$
All revealed events G_t
Batch evidence
$\overline{z_t} = \frac{\overline{\ell_t} - \mu_k}{\tilde{\sigma}_k}$
not ready
h
schematic
$C_t = \max(0, C_{t-1} + \overline{z_t} - \kappa)$
Feasible boundary
$|A_t| > 0$
$C_t \geq h$
$|B_t| \geq n_{\min}$
$b_t \geq b_{\min}$
τ_{k+1}
updated reference
Register targets and split B_t
B_{fit} (90%)
V (10%)
$b_t = \left\lfloor \frac{Sn_t}{N} \right\rfloor - s_t$
Fitting the current stage
batch
$\mathcal{L}_{\text{fit}}$
gradient step
$M_{k+1} = \mathcal{U}(M_k, B_{\text{fit}}; b_t)$
M_{k+1}
$\mu_{k+1}, \tilde{\sigma}_{k+1}$
$B \leftarrow \emptyset, C \leftarrow 0$
(b) AdaCGR-U: candidate learning and paired-utility deployment
Candidate learning
Rolling w-day fitting window
Probe V_t
later
φ_t
$b_t \geq 16$
fit b_t optimizer steps
Ω
persistent AdamW
Paired evaluation on the same V_t
θ_k
φ_t
$d_j = \ell_{\theta_k}(j) - \ell_{\varphi_t}(j)$
Accumulate paired improvement
$g_t = \frac{\overline{d_t}}{\max(\text{Std}(d), 0.05)/\sqrt{m_t}}$
0.5
$U_t = \max(0, U_{t-1} + g_t - 0.05)$
Selective deployment
$U_t \geq 0.5$
$\theta \leftarrow \varphi$
$U \leftarrow 0$
Accept: replace incumbent
Otherwise: retain incumbent
θ_k
next batch